Setup for push notification

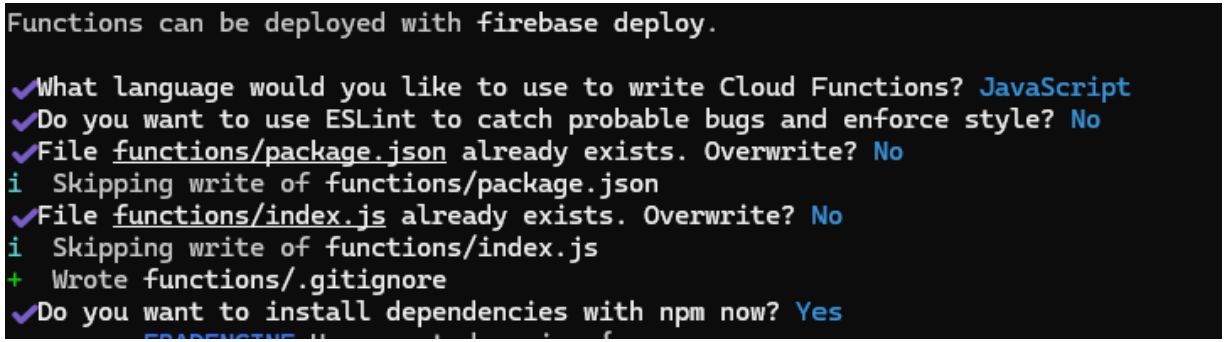
ss2026

Setup before lab

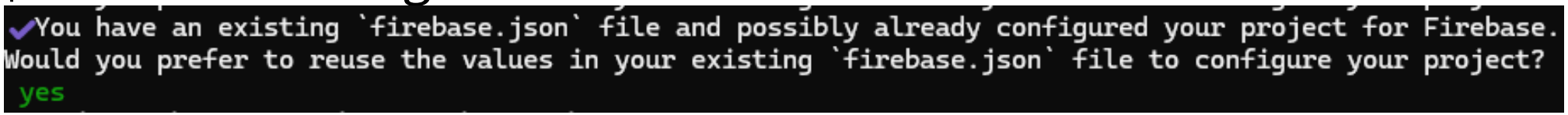
- Ensure Firestore, Storage, and Authentication are already configured and fully functional in this project. *(No need to re-clone the project).*
- If you hasn’t finished the setup teaching last weak, please follow the link to do the Firestore, Storage, and Authentication setup

Setup Functions (on your terminal)

- $ *firebase init functions*

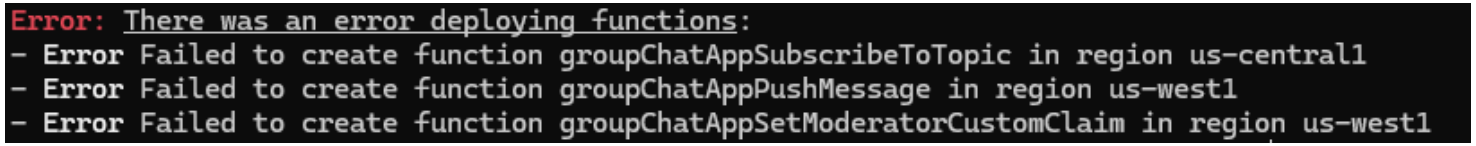


- $ *flutterfire configure*



- $ *firebase deploy --only functions*

If you face the problem :
just try several time.



Setup for push notification

- **Web (follow the below pages)**
- For web users, please download web/firebase-messaging-sw.js and set up firebase.initializeApp. Go to Firebase Console > Your Project > Project Settings > Scroll down and click on your Web App, where you can find your apiKey, appId, and other configuration values.
- In push_messanging.dart set up your vapid key. You can find your vapid key in Firebase Console > Your Project > Project Settings > Cloud Messanging > Scroll to Web configuration > click Generate Key Pair > Copy the Key Pair
- make sure to enable notification permissions by going to: System Settings > Notifications > [Your browser, e.g., Chrome or Safari].

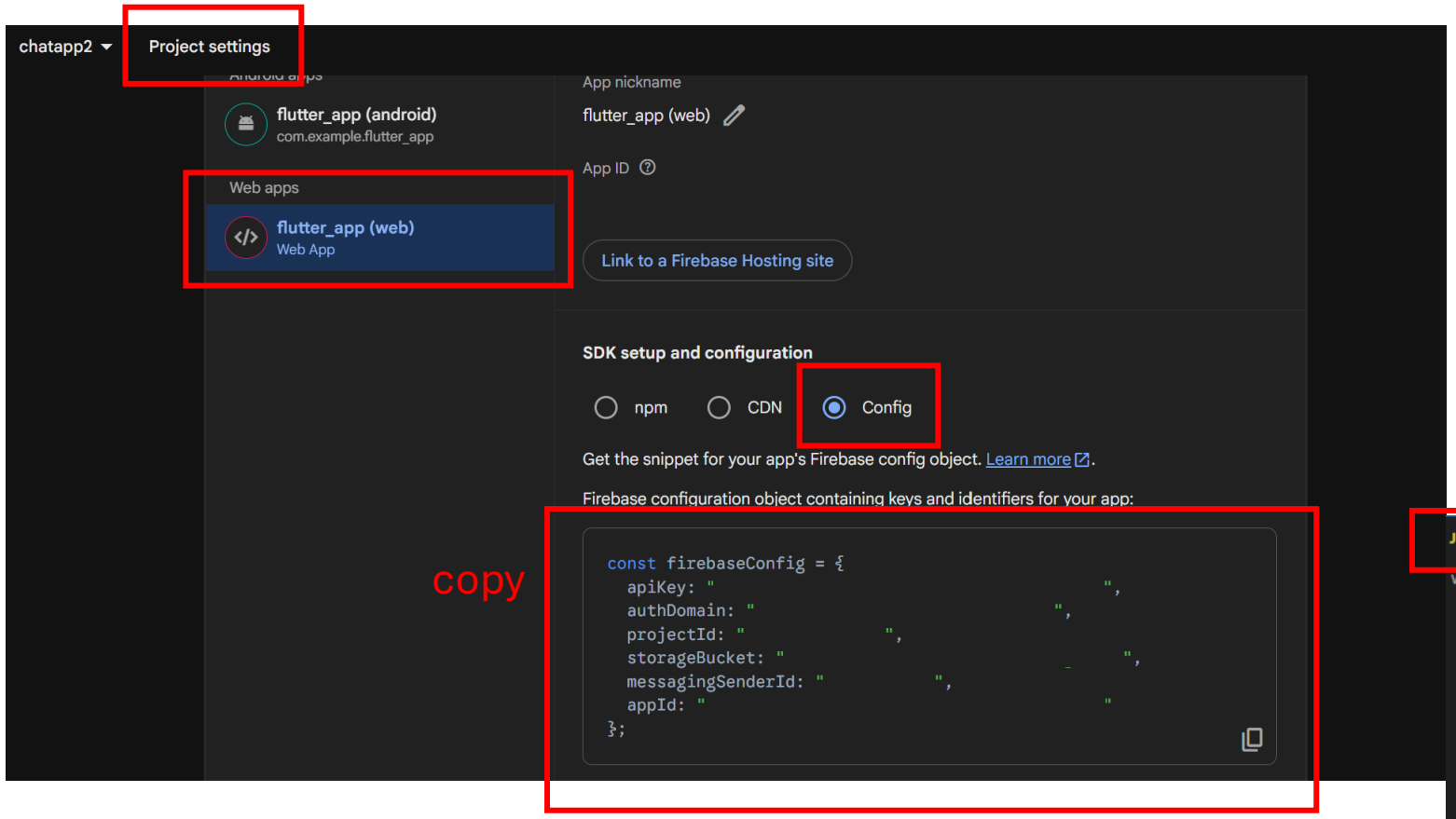
- **Android (follow the below pages) (we recommend to use web)**
  - **Step 1:** Add the SHA-256 fingerprint to your Firebase Android app.
  - **Step 2:** Refresh the console and download the updated google-services.json.
  - **Step 3:** Place the downloaded file into your local project's android/app/ directory.

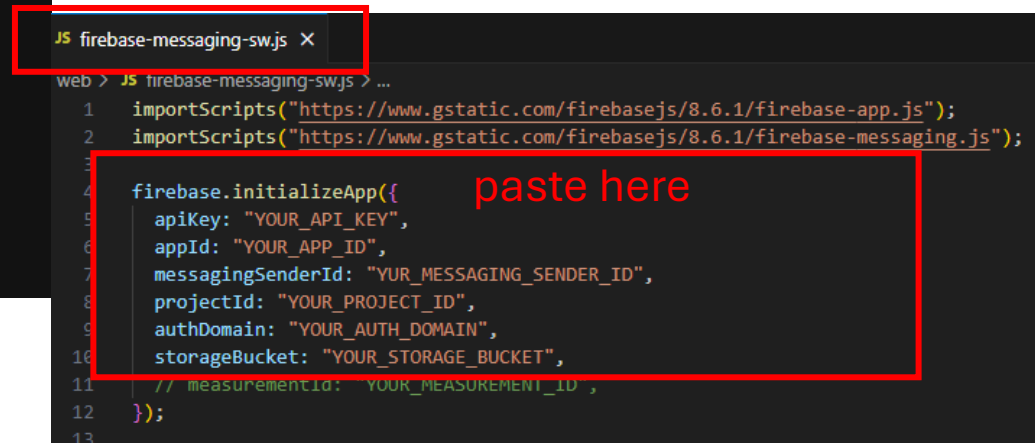
- **iOS**
- For users running the iOS simulator, receiving notifications requires APNs, which in turn requires an Apple Developer account with proper permissions.
  It is recommended to use the web simulator instead if you're unable to configure APNs.

Web setting - firebase-messaging-sw.js

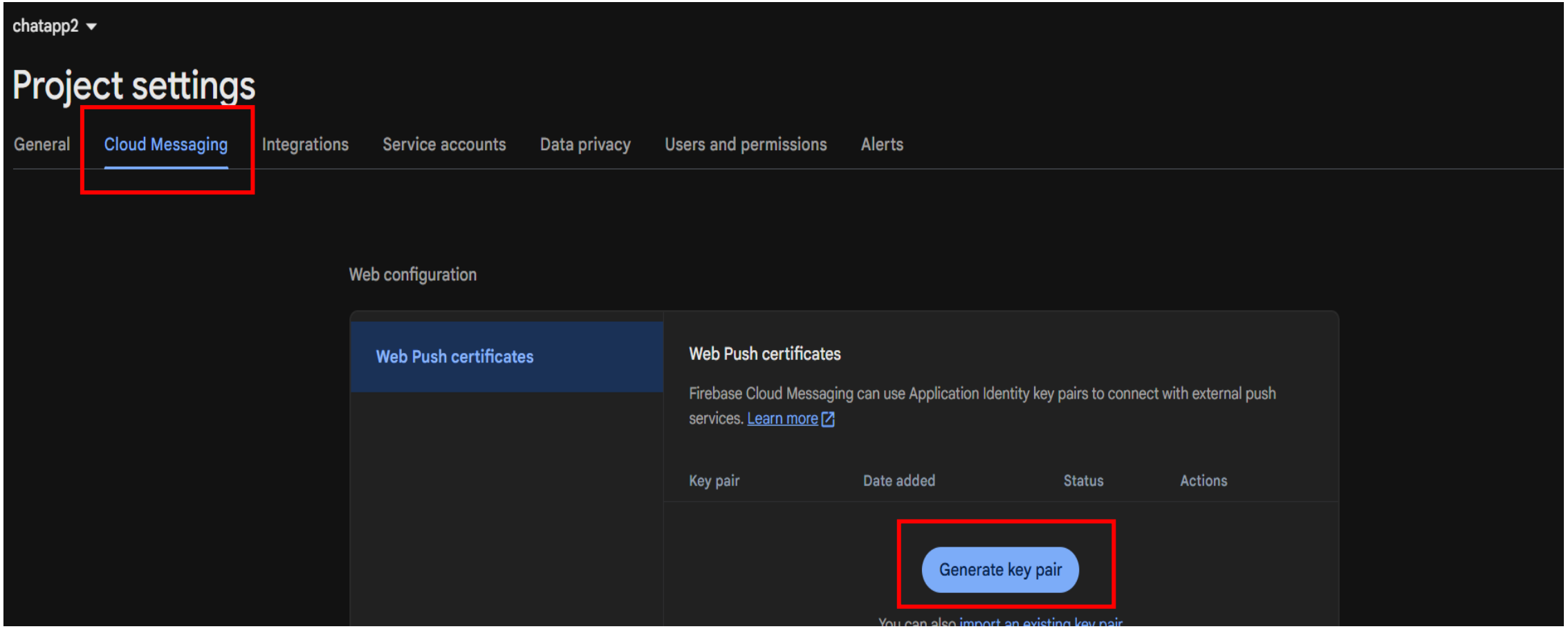
- Verify if the file web/firebase-messaging-sw.js exists.
  - If it is missing, download it from the lab webpage and place it directly into the web/firebase-messaging-sw.js.

Web setting - firebase-messaging-sw.js

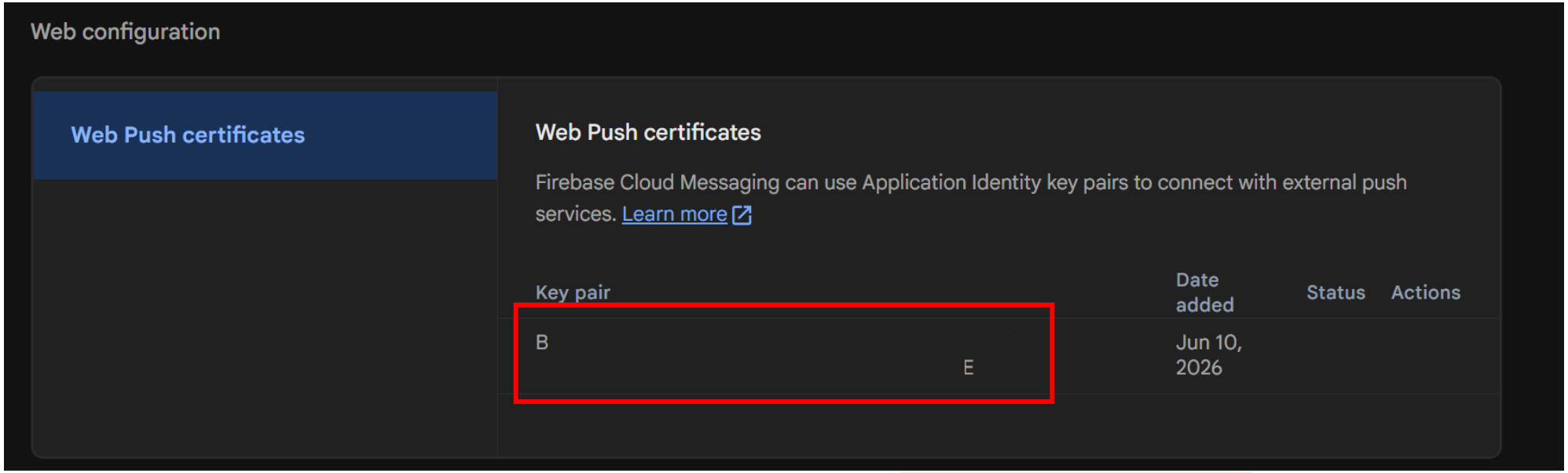


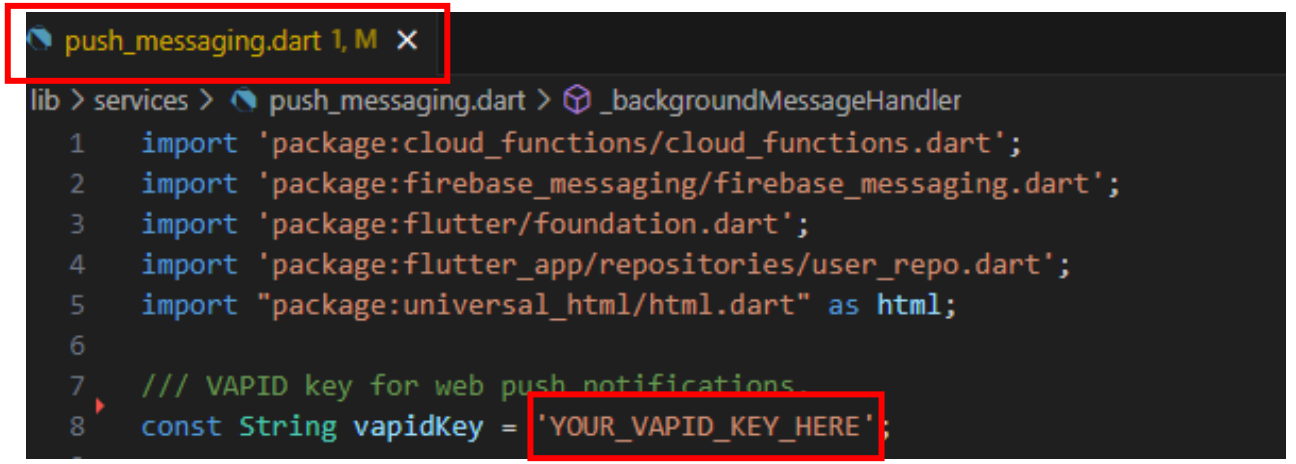


Web setting - Vapid key



Web setting – Vapid key





Web setting – Cloud Run Access Permissions

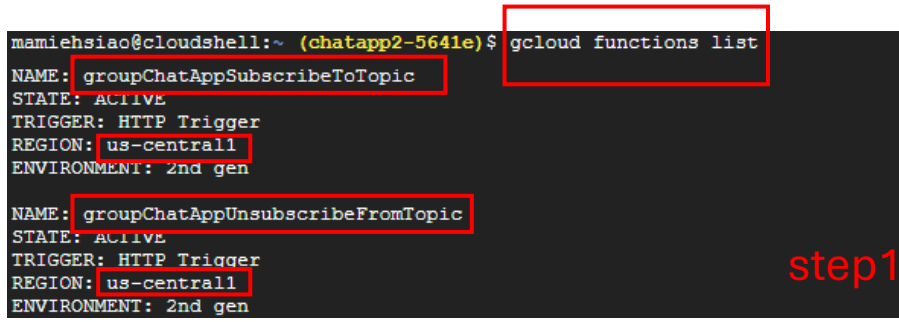
- **Step 1: Find Your Functions Region**

  Run the following command to check your deployed region on google cloud shell

  $ *gcloud functions list*

- **Step 2: Update Cloud Run Access Permissions**
  - Allow unauthenticated invocations (allUsers) for both services.
  - Note: Replace “your_region” with the region found in Step 1

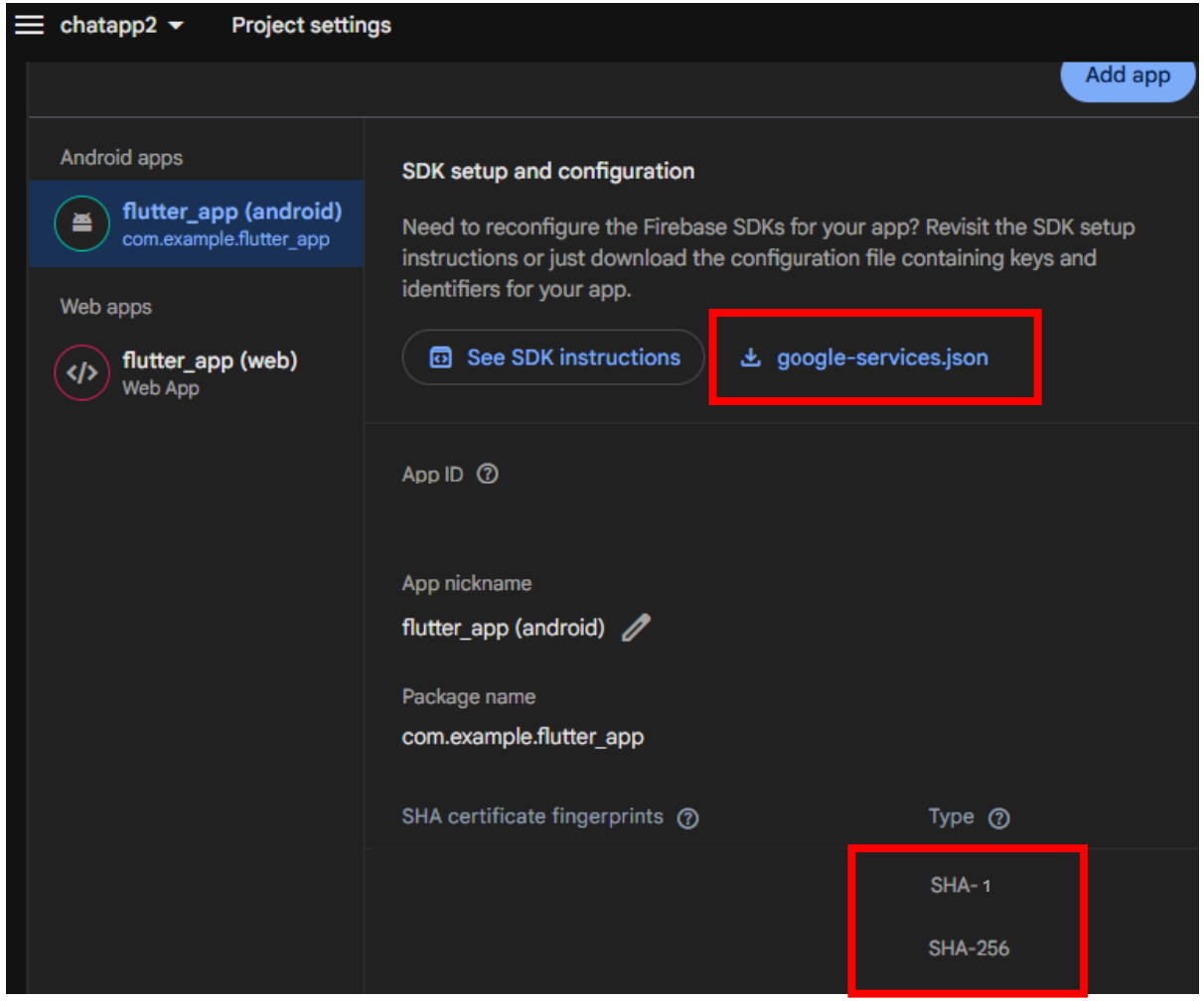
```
$ gcloud run services add-iam-policy-binding groupchatappunsubscribefromtopic \
--member="allUsers" \
--role="roles/run.invoker" \
--region="your_region"

$ gcloud run services add-iam-policy-binding groupchatappsubscribetotopic \
--member="allUsers" \
--role="roles/run.invoker" \
--region="your_region"
```



Android Emulator Setting

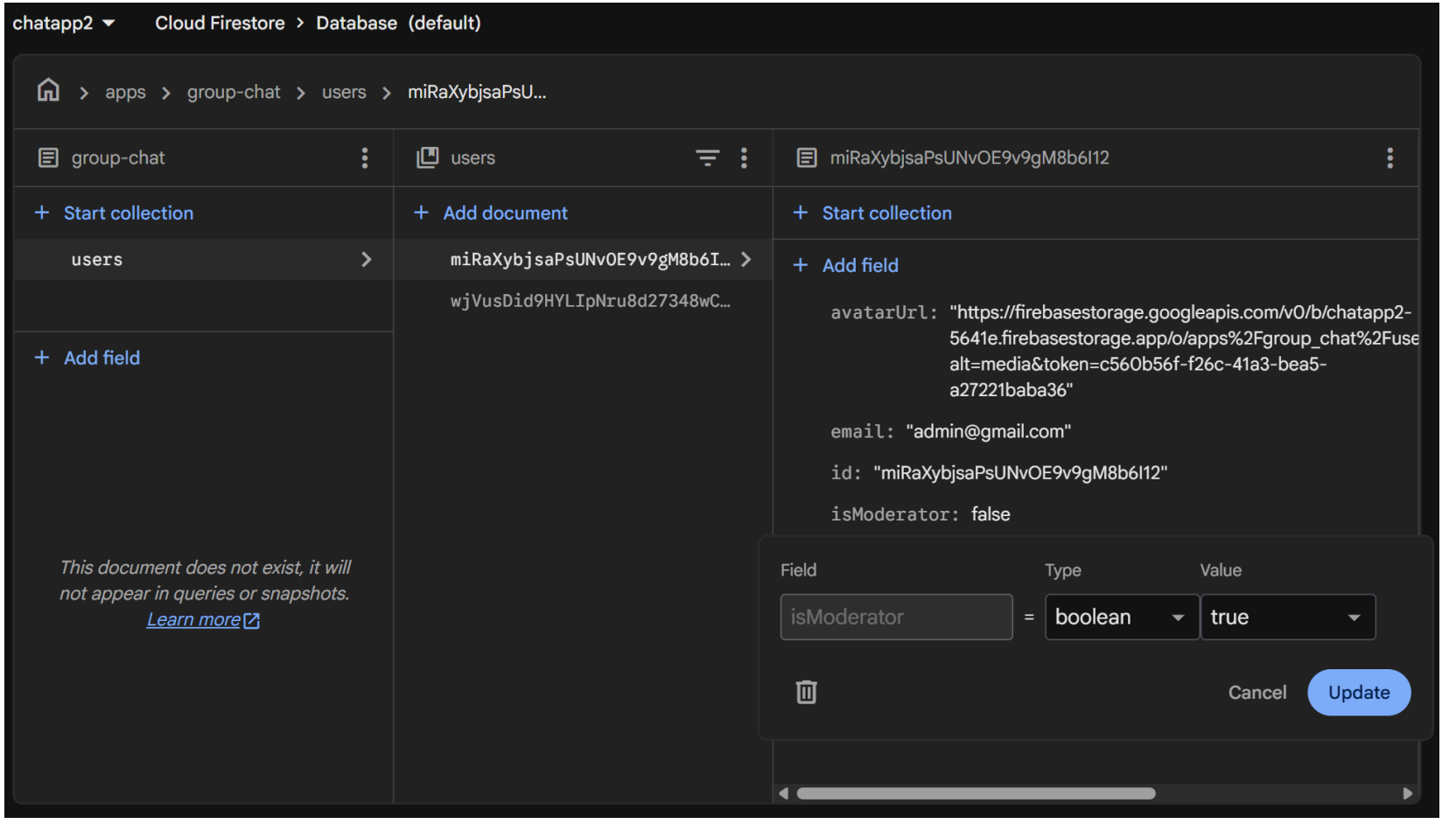
- In your cmd
  $ keytool -list -v -alias androiddebugkey -keystore "C:\Users\{your user name}\.android\debug.keystore"
  Default Password: android
- Add the SHA-256 fingerprint to your Firebase Android app.
- Step 2: Refresh the console and download the updated google-services.json.
- Step 3: Place the downloaded file into your local project's android/app

Android Emulator Setting



Let a user become moderator

- Run your project and let a user become moderator



Run Your Project

- Test Notification Functionality:
  - Run the project simultaneously in two separate windows (e.g., normal and incognito mode).
    $ flutter run -d chrome --web-port=5000
    $ flutter run -d chrome --web-port=7000
  - Send a message from User A while User B is not on the active chat page. Verify that User B receives a browser notification.
  - Example setting finishing url: https://youtu.be/2eZi1kilalw

Troubleshooting

- Troubleshooting (If notifications don't appear):
  - Ensure your computer is not on "Silent Mode" / "Do Not Disturb".
  - Check that Chrome permissions are not blocking notifications for this site.